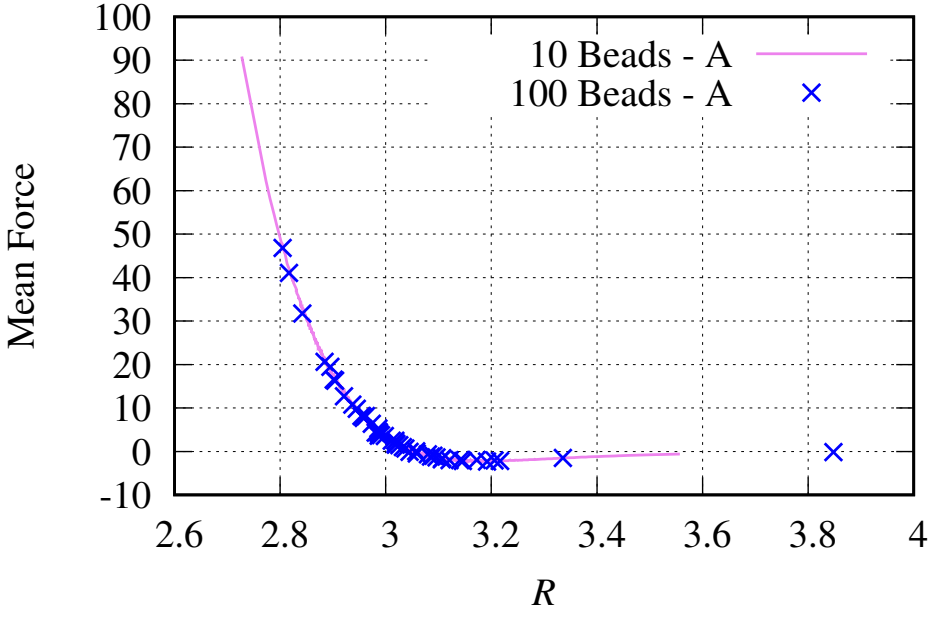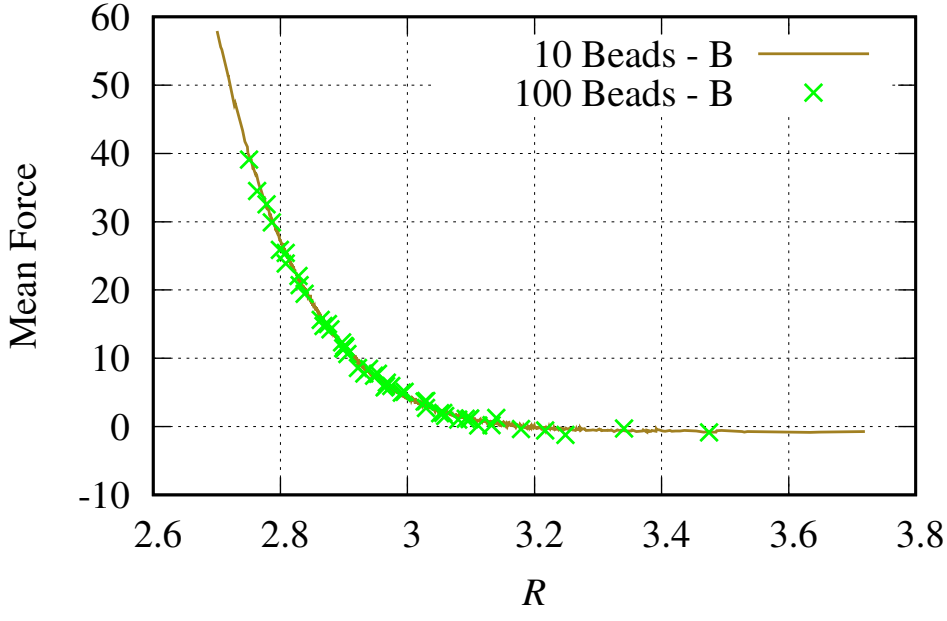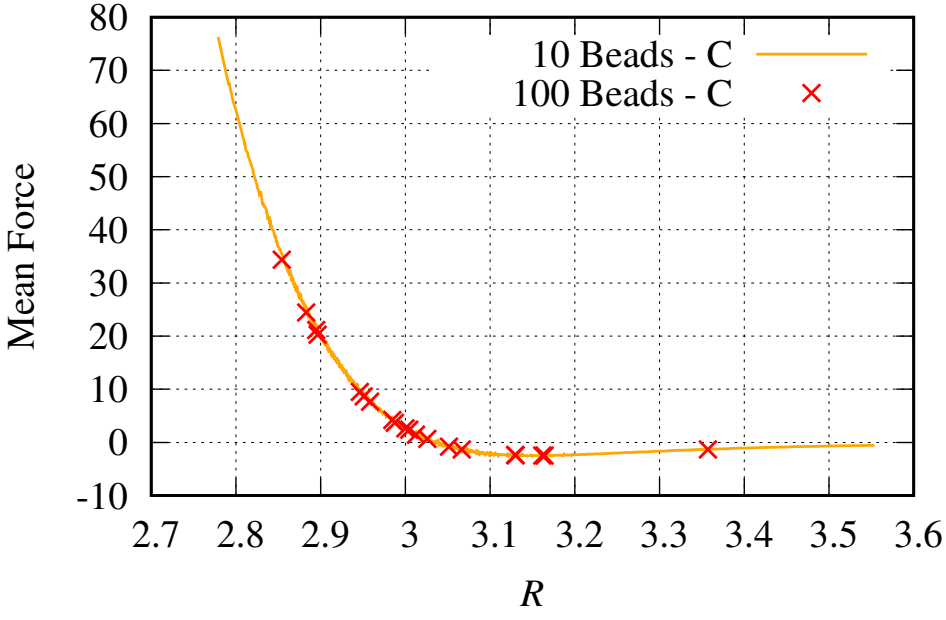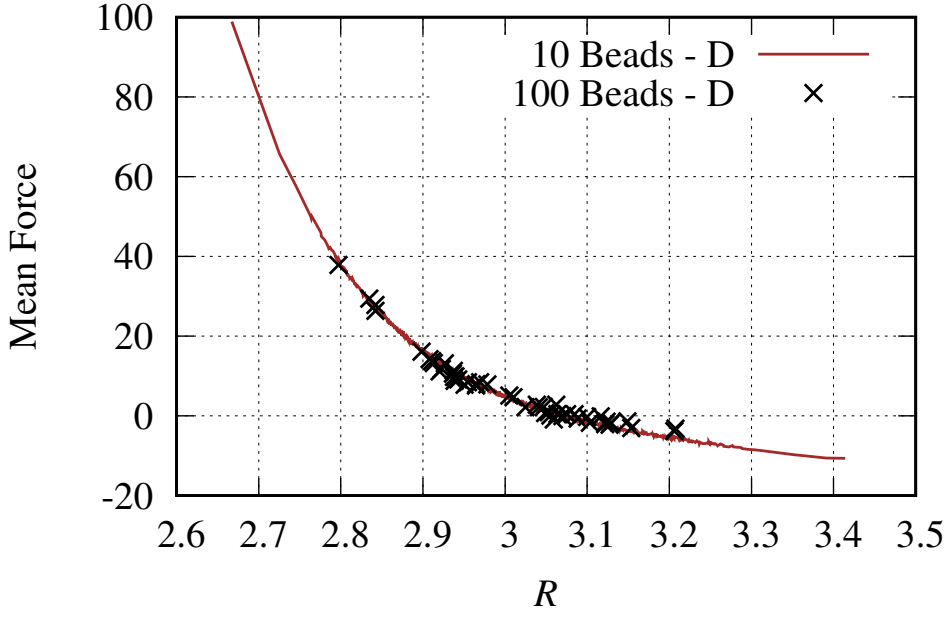System A (m-M-m) and system C (M-m-M), and system B (m-M-m) and system D (M-m-M) have the same interactions